

One-Dimensional Fatigue Modeling of Aluminum: Exact General- m, n Slip-Dependent Lattice Energy and Plastic-Slip Extension

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Abstract

This note derives, without fixing the repulsive and attractive exponents, the exact interaction energy of two parallel infinite atomic rows separated by a normal distance a and shifted by a tangential registry s . A generalized Lennard–Jones m – n pair interaction produces a shifted inhomogeneous Epstein–Hurwitz sum. Mellin transformation and Poisson summation give an exact Fourier–Bessel representation; no Taylor expansion of the energy landscape is used. The derivation is carried through to the excess slip energy, internal traction, representative-patch energy, equilibrium density, joint Smoluchowski equation, moment equations, and irreversible dissipation. An unwrapped well index converts periodic registry changes into a reduced plastic slip variable. The model can therefore represent ideal thermally activated or instability-driven single slip and its hysteresis, while quantitative plasticity of face-centered-cubic aluminum requires a crystallographic half-space energy obtained from a validated EAM or first-principles model.

1 Definitions, Counting Convention, and Scope

Let $a > 0$ be a normal atomic or interplanar spacing, s a relative tangential registry, and b the lattice repeat distance along a prescribed slip direction. The repulsive and attractive exponents remain symbolic throughout:

$$\boxed{m > n > 1.} \quad (1)$$

The lower bound on n is not a fitted choice: it is the convergence condition for an infinite one-dimensional lattice sum. The scalar s is an internal microscopic coordinate. Introducing it does not require a two-dimensional finite-element constitutive law; the surrounding finite-element mesh may still be one-, two-, or three-dimensional while the present material law uses only a selected normal component and, if plastic slip is activated, one resolved slip coordinate.

For a finite set of atoms the pair energy is counted once as

$$E_{\text{pair}} = \frac{1}{2} \sum_i \sum_{j \neq i} v(r_{ij}) = \sum_{i < j} v(r_{ij}). \quad (2)$$

The factor one half is essential when both directions from every atom are summed. By contrast, an interaction between one chosen upper atom and all lower-row atoms contains no factor one half, because every cross-row pair in that local sum occurs only once.

2 Generalized Lennard–Jones m – n Interaction

It is convenient to preserve the usual meanings of ε_{LJ} and σ_{LJ} : ε_{LJ} is the well depth and σ_{LJ} is the zero-crossing distance. The normalized general- m, n potential is

$$\boxed{v_{m,n}(r) = C_{m,n} \varepsilon_{\text{LJ}} \left[\left(\frac{\sigma_{\text{LJ}}}{r} \right)^m - \left(\frac{\sigma_{\text{LJ}}}{r} \right)^n \right], \quad C_{m,n} = \frac{m}{m-n} \left(\frac{m}{n} \right)^{n/(m-n)}.} \quad (3)$$

To derive the normalization, set $v'_{m,n}(r_e) = 0$. Differentiation gives

$$-m\sigma_{\text{LJ}}^m r_e^{-m-1} + n\sigma_{\text{LJ}}^n r_e^{-n-1} = 0, \quad (4)$$

hence

$$r_e = \sigma_{\text{LJ}} \left(\frac{m}{n} \right)^{1/(m-n)}. \quad (5)$$

Requiring $v_{m,n}(r_e) = -\varepsilon_{\text{LJ}}$ then yields the coefficient $C_{m,n}$ in Eq. (3). Eliminating σ_{LJ} by Eq. (5) gives the exactly equivalent equilibrium-distance form

$$v_{m,n}(r) = \varepsilon_{\text{LJ}} \left[\frac{n}{m-n} \left(\frac{r_e}{r} \right)^m - \frac{m}{m-n} \left(\frac{r_e}{r} \right)^n \right]. \quad (6)$$

Thus $v(r_e) = -\varepsilon_{\text{LJ}}$, $v'(r_e) = 0$, and $v''(r_e) > 0$ follow without assigning numerical values to either exponent.

2.1 Exact normal-chain energy and the two-direction issue

For N equally spaced atoms in a straight finite chain,

$$E_N(a) = \sum_{i < j} v_{m,n}((j-i)a) = \sum_{k=1}^{N-1} (N-k) v_{m,n}(ka). \quad (7)$$

The factor $N-k$ is the exact number of pairs separated by ka . Dividing by N and taking the thermodynamic limit gives the energy per atom

$$\begin{aligned} U_\infty(a) &= \lim_{N \rightarrow \infty} \frac{E_N(a)}{N} = \sum_{k=1}^{\infty} v_{m,n}(ka) \\ &= C_{m,n} \varepsilon_{\text{LJ}} \left[\left(\frac{\sigma_{\text{LJ}}}{a} \right)^m \zeta(m) - \left(\frac{\sigma_{\text{LJ}}}{a} \right)^n \zeta(n) \right]. \end{aligned} \quad (8)$$

An interior reference atom interacts in both directions, so its unhalved local interaction is $2U_\infty$. Multiplication by one half in Eq. (2) returns the thermodynamic per-atom value U_∞ . Equation (8) therefore already includes both directions with correct pair counting. A terminal atom in a semi-infinite chain has neighbors in only one direction and happens to have the same sum, but for a different geometrical reason.

The exact zero-load equilibrium of the infinite chain follows from $U'_\infty(a_0) = 0$:

$$a_0 = \sigma_{\text{LJ}} \left[\frac{m\zeta(m)}{n\zeta(n)} \right]^{1/(m-n)}. \quad (9)$$

This is a lattice equilibrium and is generally not identical to the isolated-pair distance r_e .

3 Two-Row Geometry and Direct Slip-Dependent Sum

Let $\mathbf{d}$ be a unit vector along the selected registry direction and $\mathbf{n}$ the unit normal, with $\mathbf{d} \cdot \mathbf{n} = 0$. The lower row is

$$\mathbf{R}_p^- = p\mathbf{b}\mathbf{d}, \quad p \in \mathbb{Z}, \quad (10)$$

and the vector from one upper reference atom to the p th lower atom is

$$\mathbf{r}_p(a, s) = a\mathbf{n} + (pb + s)\mathbf{d}. \quad (11)$$

Therefore

$$r_p(a, s) = \sqrt{a^2 + (pb + s)^2}. \quad (12)$$

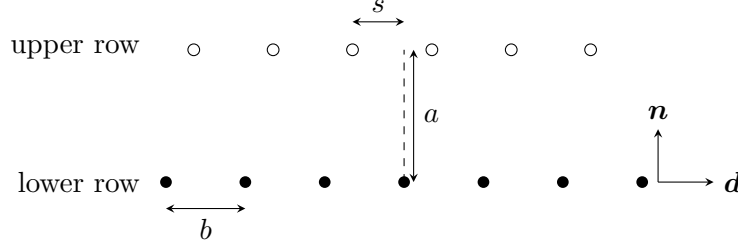


Figure 1: Reduced two-row geometry. The normal separation is a , the row repeat is b , and s is the tangential registry.

The cross-row interaction of the upper reference atom is

$$\begin{aligned} W_{m,n}(a, s) &= \sum_{p=-\infty}^{\infty} v_{m,n}(r_p) \\ &= C_{m,n}\varepsilon_{\text{LJ}} \sum_{p=-\infty}^{\infty} \left[\frac{\sigma_{\text{LJ}}^m}{[a^2 + (pb + s)^2]^{m/2}} - \frac{\sigma_{\text{LJ}}^n}{[a^2 + (pb + s)^2]^{n/2}} \right]. \end{aligned} \quad (13)$$

For large $|p|$, the two absolute terms are respectively $O(|p|^{-m})$ and $O(|p|^{-n})$. Hence Eq. (1) makes both series absolutely convergent; no cancellation between divergent repulsive and attractive sums is being used.

Set

$$\delta = \frac{s}{b}, \quad \eta = \frac{a}{b}, \quad (14)$$

and define

$$\mathcal{Z}_\nu(\delta, \eta) = \sum_{p=-\infty}^{\infty} [(p + \delta)^2 + \eta^2]^{-\nu}, \quad \eta > 0, \quad \text{Re } \nu > \frac{1}{2}. \quad (15)$$

Then

$$W_{m,n}(a, s) = C_{m,n}\varepsilon_{\text{LJ}} \left[\left(\frac{\sigma_{\text{LJ}}}{b} \right)^m \mathcal{Z}_{m/2}(\delta, \eta) - \left(\frac{\sigma_{\text{LJ}}}{b} \right)^n \mathcal{Z}_{n/2}(\delta, \eta) \right]. \quad (16)$$

Because reindexing $p \mapsto p - 1$ leaves the sum unchanged, $W_{m,n}(a, s + b) = W_{m,n}(a, s)$ exactly.

4 Exact Mellin–Poisson–Bessel Derivation

4.1 Mellin representation and interchange of sum and integral

For $x > 0$ and $\text{Re } \nu > 0$,

$$x^{-\nu} = \frac{1}{\Gamma(\nu)} \int_0^\infty t^{\nu-1} e^{-xt} dt. \quad (17)$$

Substitution into every positive term of Eq. (15) gives

$$\begin{aligned}\mathcal{Z}_\nu(\delta, \eta) &= \frac{1}{\Gamma(\nu)} \sum_{p \in \mathbb{Z}} \int_0^\infty t^{\nu-1} e^{-t[(p+\delta)^2 + \eta^2]} dt \\ &= \frac{1}{\Gamma(\nu)} \int_0^\infty t^{\nu-1} e^{-\eta^2 t} \sum_{p \in \mathbb{Z}} e^{-t(p+\delta)^2} dt.\end{aligned}\quad (18)$$

For real $\nu > 1/2$ the integrand is nonnegative, so Tonelli's theorem justifies the interchange. For complex ν in the same half-plane, absolute convergence gives the corresponding Fubini justification.

4.2 Shifted Gaussian under Poisson summation

Use the Fourier-transform convention

$$\widehat{g}(\ell) = \int_{-\infty}^\infty g(x) e^{-2\pi i \ell x} dx. \quad (19)$$

For $g(x) = e^{-t(x+\delta)^2}$, the change of variable $y = x + \delta$ gives

$$\begin{aligned}\widehat{g}(\ell) &= e^{2\pi i \ell \delta} \int_{-\infty}^\infty e^{-ty^2} e^{-2\pi i \ell y} dy \\ &= \sqrt{\frac{\pi}{t}} e^{-\pi^2 \ell^2 / t} e^{2\pi i \ell \delta}.\end{aligned}\quad (20)$$

Poisson summation $\sum_p g(p) = \sum_\ell \widehat{g}(\ell)$ therefore gives

$$\boxed{\sum_{p \in \mathbb{Z}} e^{-t(p+\delta)^2} = \sqrt{\frac{\pi}{t}} \sum_{\ell \in \mathbb{Z}} e^{-\pi^2 \ell^2 / t} e^{2\pi i \ell \delta}.} \quad (21)$$

Inserting Eq. (21) into Eq. (18) yields

$$\mathcal{Z}_\nu = \frac{\sqrt{\pi}}{\Gamma(\nu)} \sum_{\ell \in \mathbb{Z}} e^{2\pi i \ell \delta} \int_0^\infty t^{\nu-3/2} \exp\left(-\eta^2 t - \frac{\pi^2 \ell^2}{t}\right) dt. \quad (22)$$

4.3 Zero and nonzero reciprocal modes

For $\ell = 0$, set $u = \eta^2 t$:

$$\begin{aligned}I_0 &= \int_0^\infty t^{\nu-3/2} e^{-\eta^2 t} dt \\ &= \eta^{1-2\nu} \int_0^\infty u^{\nu-3/2} e^{-u} du = \Gamma\left(\nu - \frac{1}{2}\right) \eta^{1-2\nu}.\end{aligned}\quad (23)$$

For $\ell \neq 0$, apply the exact identity

$$\int_0^\infty t^{\mu-1} e^{-\beta t - \gamma/t} dt = 2 \left(\frac{\gamma}{\beta}\right)^{\mu/2} K_\mu(2\sqrt{\beta\gamma}), \quad (24)$$

with

$$\mu = \nu - \frac{1}{2}, \quad \beta = \eta^2, \quad \gamma = \pi^2 \ell^2. \quad (25)$$

It follows directly that

$$I_\ell = 2 \left(\frac{\pi|\ell|}{\eta} \right)^{\nu-1/2} K_{\nu-1/2}(2\pi|\ell|\eta). \quad (26)$$

Because $I_{-\ell} = I_\ell$ and $e^{2\pi i\ell\delta} + e^{-2\pi i\ell\delta} = 2\cos(2\pi\ell\delta)$, Eq. (22) becomes

$$\mathcal{Z}_\nu(\delta, \eta) = \frac{\sqrt{\pi}}{\Gamma(\nu)} \left[\Gamma\left(\nu - \frac{1}{2}\right) \eta^{1-2\nu} + 4 \sum_{\ell=1}^{\infty} \cos(2\pi\ell\delta) \left(\frac{\pi\ell}{\eta} \right)^{\nu-1/2} K_{\nu-1/2}(2\pi\ell\eta) \right]. \quad (27)$$

Equation (27) is an identity, not a long-wavelength, small-slip, or harmonic approximation. Its reciprocal terms converge exponentially because $K_\alpha(x) \sim \sqrt{\pi/(2x)}e^{-x}$ as $x \rightarrow \infty$.

5 Closed Full Energy for Symbolic m, n

For any exponent $q > 1$, define

$$\mathcal{B}_q(\delta, \eta) := \frac{\sqrt{\pi}}{\Gamma(q/2)} \left[\Gamma\left(\frac{q-1}{2}\right) \eta^{1-q} + 4 \sum_{\ell=1}^{\infty} \cos(2\pi\ell\delta) \left(\frac{\pi\ell}{\eta} \right)^{(q-1)/2} K_{(q-1)/2}(2\pi\ell\eta) \right]. \quad (28)$$

Thus $\mathcal{B}_q = \mathcal{Z}_{q/2}$. Substitution of Eq. (28) into the generalized pair energy gives the requested full, exact, exponent-symbolic expression:

$$W_{m,n}(a, s) = C_{m,n}\varepsilon_{\text{LJ}} \left\{ \left(\frac{\sigma_{\text{LJ}}}{b} \right)^m \frac{\sqrt{\pi}}{\Gamma(m/2)} \left[\Gamma\left(\frac{m-1}{2}\right) \eta^{1-m} + 4 \sum_{\ell=1}^{\infty} \cos(2\pi\ell\delta) \left(\frac{\pi\ell}{\eta} \right)^{(m-1)/2} K_{(m-1)/2}(2\pi\ell\eta) \right] - \left(\frac{\sigma_{\text{LJ}}}{b} \right)^n \frac{\sqrt{\pi}}{\Gamma(n/2)} \left[\Gamma\left(\frac{n-1}{2}\right) \eta^{1-n} + 4 \sum_{\ell=1}^{\infty} \cos(2\pi\ell\delta) \left(\frac{\pi\ell}{\eta} \right)^{(n-1)/2} K_{(n-1)/2}(2\pi\ell\eta) \right] \right\}, \quad (29)$$

$$\delta = \frac{s}{b}, \quad \eta = \frac{a}{b}.$$

Every factor multiplying ε_{LJ} in Eq. (29) is dimensionless, so $[W_{m,n}] = \text{energy}$. No exponent has been assigned a numerical value.

6 Excess Slip Energy, Traction, and Exact Barrier

Let s_0 be a chosen perfect registry, $\delta_0 = s_0/b$, and let A_{at} denote the slip-plane area represented by one reference atom. Define

$$\Delta W_{m,n}(a, s) = W_{m,n}(a, s) - W_{m,n}(a, s_0), \quad \gamma_{m,n}(a, s) = \frac{\Delta W_{m,n}(a, s)}{A_{\text{at}}}. \quad (30)$$

The registry-independent zero Fourier mode cancels. For $q > 1$ define

$$\Delta \mathcal{B}_q(\delta, \delta_0; \eta) = \frac{4\sqrt{\pi}}{\Gamma(q/2)} \sum_{\ell=1}^{\infty} [\cos(2\pi\ell\delta) - \cos(2\pi\ell\delta_0)] \times \left(\frac{\pi\ell}{\eta}\right)^{(q-1)/2} K_{(q-1)/2}(2\pi\ell\eta). \quad (31)$$

The complete generalized slip energy is therefore

$$\gamma_{m,n}(a, s) = \frac{C_{m,n}\varepsilon_{\text{LJ}}}{A_{\text{at}}} \left[\left(\frac{\sigma_{\text{LJ}}}{b}\right)^m \Delta \mathcal{B}_m - \left(\frac{\sigma_{\text{LJ}}}{b}\right)^n \Delta \mathcal{B}_n \right]. \quad (32)$$

The internal resisting traction is

$$\tau_{\text{int}}(a, s) = \frac{\partial \gamma_{m,n}}{\partial s} = \frac{C_{m,n}\varepsilon_{\text{LJ}}}{A_{\text{at}}b} \left[\left(\frac{\sigma_{\text{LJ}}}{b}\right)^m \partial_{\delta} \mathcal{B}_m - \left(\frac{\sigma_{\text{LJ}}}{b}\right)^n \partial_{\delta} \mathcal{B}_n \right], \quad (33)$$

where the exact derivative for either $q = m$ or $q = n$ is

$$\partial_{\delta} \mathcal{B}_q(\delta, \eta) = -\frac{8\pi\sqrt{\pi}}{\Gamma(q/2)} \sum_{\ell=1}^{\infty} \ell \sin(2\pi\ell\delta) \left(\frac{\pi\ell}{\eta}\right)^{(q-1)/2} K_{(q-1)/2}(2\pi\ell\eta). \quad (34)$$

Because $[\gamma] = \text{energy/area}$ and $[s] = \text{length}$, $[\tau_{\text{int}}] = \text{force/area}$. At fixed a , the ideal lattice resistance is

$$\tau_{\text{ideal}}(a) = \max_{s \in [s_0, s_0+b)} |\tau_{\text{int}}(a, s)|. \quad (35)$$

Under a resolved external shear $\tau(t)$, the tilted energy of a patch of area A_0 is

$$\psi(a, s, t) = A_0 \gamma_{m,n}(a, s) - \tau(t) A_0 (s - s_0). \quad (36)$$

A stable registry s_z and a neighboring saddle $s_z^{\ddagger}$ are obtained from the full energy, not from a local polynomial:

$$\partial_s \gamma_{m,n}(a, s_z) = \tau, \quad \partial_s^2 \gamma_{m,n}(a, s_z) > 0, \quad (37)$$

$$\partial_s \gamma_{m,n}(a, s_z^{\ddagger}) = \tau, \quad \partial_s^2 \gamma_{m,n}(a, s_z^{\ddagger}) < 0. \quad (38)$$

The exact forward or backward activation barrier is

$$\Delta G_{z,\pm}^{\ddagger}(a, \tau) = A_0 \left[\gamma_{m,n}(a, s_{z,\pm}^{\ddagger}) - \gamma_{m,n}(a, s_z) - \tau(s_{z,\pm}^{\ddagger} - s_z) \right]. \quad (39)$$

Loss of metastability occurs when a minimum and saddle coalesce, equivalently

$$\partial_s \gamma_{m,n} = \tau, \quad \partial_s^2 \gamma_{m,n} = 0. \quad (40)$$

7 Full Representative-Patch Energy Under External Loading

Let a representative patch have area A_0 and contain $N_A = A_0/A_{\text{at}}$ equivalent atomic columns. Using the normal chain energy as the reference and adding only the excess registry energy prevents double counting. The reduced full potential is

$$\begin{aligned} \Phi(a, s, t) = N_A C_{m,n} \varepsilon_{\text{LJ}} & \left[\left(\frac{\sigma_{\text{LJ}}}{a} \right)^m \zeta(m) - \left(\frac{\sigma_{\text{LJ}}}{a} \right)^n \zeta(n) \right] \\ & + A_0 \gamma_{m,n}(a, s) - \sigma(t) A_0 (a - a_0) - \tau(t) A_0 (s - s_0). \end{aligned} \quad (41)$$

All four terms have units of energy. The stress-to-atomic-force conversion is not arbitrary: $F_n = \sigma A_0$ and $F_s = \tau A_0$ follow from the definition of traction on the represented area. For uniaxial loading and one specified slip system,

$$\tau(t) = \mathcal{M} \sigma(t), \quad (42)$$

where $\mathcal{M}$ is the crystallographic resolved-shear projection. In the strict normal-only branch one sets $s = s_0$ and $\tau = 0$, and Eq. (41) reduces to a scalar normal-spacing energy.

At fixed load and inside a bounded bonded basin Ω_b , statistical mechanics gives

$$P_{\text{eq}}(a, s \mid \sigma, \tau) = \frac{1}{\mathcal{Z}_{\text{conf}}} \exp[-\beta_T \Phi(a, s)], \quad \beta_T = (k_B T)^{-1}, \quad (43)$$

with

$$\mathcal{Z}_{\text{conf}} = \iint_{\Omega_b} e^{-\beta_T \Phi(a, s)} \, da \, ds. \quad (44)$$

Under dead tensile stress, $-\sigma A_0 a$ makes the potential unbounded below as $a \rightarrow \infty$. Consequently there is no globally normalizable stress-controlled equilibrium on $0 < a < \infty$. The physically relevant object before fracture is a metastable bonded-basin density with an absorbing escape boundary, or a displacement-controlled finite-system ensemble. This distinction is necessary when the spacing tail is used as a crack-initiation measure.

The normal part of the reduced energy is

$$\Phi_n(a, t) = N_A U_{\infty}(a) - \sigma(t) A_0 (a - a_0). \quad (45)$$

At fixed tensile stress below loss of stability, the outer barrier $a^{\ddagger}(\sigma)$ is defined without an arbitrary critical distance by

$$N_A U'_{\infty}(a^{\ddagger}) = \sigma A_0, \quad U''_{\infty}(a^{\ddagger}) < 0. \quad (46)$$

The stable spacing obeys the same force balance with positive curvature. The two stationary points coalesce at the exact lattice inflection spacing

$$a_{\text{inf}} = \sigma_{\text{LJ}} \left[\frac{m(m+1)\zeta(m)}{n(n+1)\zeta(n)} \right]^{1/(m-n)}, \quad \sigma_c = \frac{N_A}{A_0} U'_{\infty}(a_{\text{inf}}). \quad (47)$$

Thus a physically derived spacing-tail measure is

$$p_{\text{tail}}(t; \sigma) = \int_{a^{\ddagger}(\sigma)}^{\infty} \int P(a, s, t) \, ds \, da, \quad (48)$$

or, with an absorbing boundary at $a^{\ddagger}$, the accumulated outward probability flux. This connects crack initiation to the full loss-of-stability landscape rather than to a fitted a_c .

8 Nonequilibrium Probability Evolution and Hysteresis

Assume that unresolved fast degrees of freedom act as a thermal bath, inertia of a and s has relaxed, and the minimal mobility tensor is diagonal. The overdamped Langevin equations are

$$\dot{a} = -M_a \partial_a \Phi + \sqrt{2k_B T M_a} \xi_a(t), \quad (49)$$

$$\dot{s} = -M_s \partial_s \Phi + \sqrt{2k_B T M_s} \xi_s(t), \quad (50)$$

with independent unit white noises. Fluctuation–dissipation fixes the noise amplitudes; they are not independent fitting terms. The equivalent joint Smoluchowski equation is

$$\boxed{\frac{\partial P}{\partial t} = -\frac{\partial J_a}{\partial a} - \frac{\partial J_s}{\partial s}}, \quad (51)$$

where

$$J_a = -M_a (P \partial_a \Phi + k_B T \partial_a P), \quad (52)$$

$$J_s = -M_s (P \partial_s \Phi + k_B T \partial_s P). \quad (53)$$

Here $[M_a] = [M_s] = \text{length}^2/(\text{energy time})$, $[P] = \text{length}^{-2}$, and $[J_a] = [J_s] = (\text{length time})^{-1}$, so every term in Eq. (51) has units $(\text{length}^2 \text{ time})^{-1}$.

For a closed intact domain with no-flux boundaries,

$$\boxed{\iint_{\Omega_b} P(a, s, t) da ds = 1.} \quad (54)$$

With an absorbing fracture boundary, the integral is the intact survival probability $S(t) \leq 1$, and $P_{\text{crack}}(t) = 1 - S(t)$. The moments below are unconditioned moments of the intact population; conditional moments among surviving states are obtained by division by $S(t)$. The principal moments are

$$\bar{a}(t) = \iint_{\Omega_b} a P da ds, \quad (55)$$

$$\bar{s}(t) = \iint_{\Omega_b} s P da ds, \quad (56)$$

$$\text{Var}[a](t) = \iint_{\Omega_b} [a - \bar{a}(t)]^2 P da ds, \quad (57)$$

$$\bar{E}(t) = \iint_{\Omega_b} \Phi(a, s, t) P da ds. \quad (58)$$

For boundaries on which P vanishes, or on an unbounded domain with sufficiently rapid decay, integration by parts gives the exact mean evolution

$$\frac{d\bar{a}}{dt} = \iint J_a da ds = -M_a \langle \partial_a \Phi \rangle, \quad (59)$$

$$\frac{d\langle a^2 \rangle}{dt} = -2M_a \langle a \partial_a \Phi \rangle + 2M_a k_B T. \quad (60)$$

These equations are exact but not generally closed, because the nonlinear full potential couples higher moments. Closing them by a quadratic Taylor expansion would discard precisely the energy tails and barriers of interest, so the numerical model should advance P itself.

The nonequilibrium free-energy functional is

$$\mathcal{F}[P, t] = \iint_{\Omega_b} \left[\Phi P + k_B T P \ln \left(\frac{P}{P_*} \right) \right] da ds, \quad (61)$$

where P_* only makes the logarithm dimensionless. Using Eqs. (51)–(53) gives

$$\frac{d\mathcal{F}}{dt} = \langle \partial_t \Phi \rangle - \dot{D}_{\text{irr}}, \quad \dot{D}_{\text{irr}} = \iint_{\Omega_b} \left[\frac{J_a^2}{M_a P} + \frac{J_s^2}{M_s P} \right] da ds \geq 0. \quad (62)$$

Thus the accumulated irreversible energy is

$$D_{\text{irr}}(t) = \int_0^t \dot{D}_{\text{irr}}(t') dt'. \quad (63)$$

This is a physics-based hysteresis accumulation equation. Under periodic loading, finite mobilities prevent P from instantaneously attaining Eq. (43); the nonzero probability currents in Eq. (62) make loading and unloading follow different paths. In a periodic steady state the free-energy change over one cycle is zero, and the loop work satisfies

$$\oint [\sigma A_0 d\bar{a} + \tau A_0 d\bar{s}] = \int_{\text{cycle}} \dot{D}_{\text{irr}} dt \geq 0. \quad (64)$$

If the density were forced to remain at instantaneous equilibrium at every load, the quasistatic closed path would be reversible and this rate-dependent hysteresis would disappear. Permanent plastic strain requires the additional interwell bookkeeping described next.

9 Can the Model Represent Plastic Deformation?

9.1 Reduced ideal slip: yes, if the registry is unwrapped

The energy $\gamma_{m,n}(a, s)$ is periodic, so storing only $s \bmod b$ loses the number of completed lattice translations. Write

$$s(t) = z(t)b + \tilde{s}(t), \quad z \in \mathbb{Z}, \quad 0 \leq \tilde{s} < b. \quad (65)$$

Motion of $\tilde{s}$ within one well is predominantly recoverable. A change in the integer z is an irreversible slip event at the reduced-model level. If h_{slip} is the thickness over which the displacement discontinuity is homogenized, the plastic shear strain is

$$\gamma_p(t) = \frac{b}{h_{\text{slip}}} \langle z(t) \rangle, \quad \varepsilon_p(t) = \mathcal{M} \gamma_p(t) \quad (66)$$

for a single slip system under a uniaxial projection $\mathcal{M}$. This supplies a permanent strain even after $\tilde{s}$ returns near its original position.

9.2 Exact fixed-load first-passage rate, without a local energy expansion

At fixed a and frozen load, let s_L be a reflecting boundary on the left of a well and s_R an absorbing boundary beyond the selected saddle. The mean first-passage time $T_{\text{FP}}(s)$ of the slip Smoluchowski process obeys the backward equation

$$-1 = -M_s \psi'(s) T'_{\text{FP}}(s) + M_s k_B T T''_{\text{FP}}(s), \quad T'_{\text{FP}}(s_L) = 0, \quad T_{\text{FP}}(s_R) = 0. \quad (67)$$

Multiplying by $e^{-\beta_T \psi}$, integrating once from s_L to an intermediate coordinate, and then applying the absorbing condition gives the exact quadrature

$$T_{\text{FP}}(s) = \frac{1}{M_s k_B T} \int_s^{s_R} e^{\beta_T \psi(y)} \left[\int_{s_L}^y e^{-\beta_T \psi(x)} dx \right] dy. \quad (68)$$

The frozen-load effective escape rate can be defined by

$$k_{z,+}^{\text{FP}}(a, \tau) = [T_{\text{FP}}(s_z)]^{-1}, \quad (69)$$

with the boundaries reversed for the backward rate. Equation (68) evaluates the full energy ψ and makes no quadratic or Taylor approximation. A time-local master equation using this rate additionally assumes separation between intrawell equilibration, interwell escape, and load variation. If that separation fails, the time-dependent Smoluchowski equation (51) is the governing equation and should be solved directly.

For comparison only, a high-barrier asymptotic reduction of the exact first-passage result is the overdamped Kramers formula

$$k_{z,\pm}(a, t) = \frac{M_s}{2\pi} \sqrt{\psi''(a, s_z, t) \left| \psi''(a, s_{z,\pm}^\ddagger, t) \right|} \exp[-\beta_T \Delta G_{z,\pm}^\ddagger(a, \tau(t))]. \quad (70)$$

The barrier in this expression is still evaluated from the full energy, but the curvature prefactor is a local harmonic approximation. It is therefore optional and is not used to define the fundamental evolution law in this work.

With well probabilities $p_z(a, t)$ and either exact first-passage rates under time-scale separation or explicitly declared asymptotic rates, a discrete-continuous alternative is

$$\partial_t p_z = -\partial_a J_{a,z} + k_{z-1,+} p_{z-1} + k_{z+1,-} p_{z+1} - (k_{z,+} + k_{z,-}) p_z. \quad (71)$$

It gives

$$\frac{d\langle z \rangle}{dt} = \sum_z \int [k_{z,+}(a, t) - k_{z,-}(a, t)] p_z(a, t) da, \quad (72)$$

and the plastic contribution to the loop dissipation is $\oint \tau d\gamma_p$. Interwell jumps, finite kinetic relaxation, and an evolving population among z states are the precise mechanisms that keep unloading from retracing loading.

9.3 What the reduced model does not yet contain

The two-row central-force model can represent an ideal periodic lattice resistance, thermally activated registry jumps, permanent single-slip strain, and dissipative hysteresis. It does not by itself contain dislocation lines, their elastic fields, multiplication, forest hardening, backstress, junctions, grain boundaries, or multiple FCC slip systems. Accordingly:

- If s is suppressed and only normal stress and a are retained, the model describes elastic opening, metastable decohesion, and damage-tail evolution, not crystallographic plasticity.
- If one s and its well index z are retained, it is a reduced ideal single-slip plasticity model suitable for mechanism studies and mathematical hysteresis.
- Quantitative cyclic plasticity of aluminum requires a validated crystalline energy surface and additional internal variables for dislocation storage and hardening.

10 EAM Extension of the Probability Framework

For a metallic crystal, the pair interaction can be replaced by the embedded-atom method [1]. Its total energy is

$$E_{\text{EAM}}(a, s) = \frac{1}{2} \sum_i \sum_{j \neq i} \phi(r_{ij}(a, s)) + \sum_i F(\rho_i(a, s)), \quad \rho_i(a, s) = \sum_{j \neq i} f(r_{ij}(a, s)). \quad (73)$$

For the proposed density kernel one may use

$$f(r) = f_0 \exp \left[-\beta_\rho \left(\frac{r}{r_e} - 1 \right) \right], \quad (74)$$

where β_ρ is dimensionless and is distinct from the thermodynamic inverse temperature β_T . The embedding term is not obtained by inserting $f(r)$ as another pair potential. The neighbor contributions must first be summed to form ρ_i , and the nonlinear function $F(\rho_i)$ is then evaluated. This order is what makes EAM many-body.

The corresponding generalized stacking-fault energy for a periodic interface cell is

$$\gamma_{\text{EAM}}(a, s) = \frac{E_{\text{EAM}}(a, s) - E_{\text{EAM}}(a, s_0)}{A_{\text{cell}}}. \quad (75)$$

Equations (41)–(71) remain structurally unchanged after replacing $\gamma_{m,n}$ by γ_{EAM} . The exponential density sum and a general nonlinear F do not reduce to the same m, n Epstein sum; they should be evaluated from the exact neighbor geometry rather than forced into an incorrect zeta form. A validated aluminum EAM can reproduce metallic cohesive energetics, stacking-fault barriers, and atomistic dislocation motion far more realistically than pair LJ, but a single collective s still represents only idealized single slip.

11 Crystalline Half-Space Generalization

Let $\mathcal{U}$ and $\mathcal{L}$ be upper and lower crystalline half-spaces separated by a chosen plane. For pair interactions the exact excess energy per periodic interface cell is

$$\gamma_{\text{GSF}}(a, s) = \frac{1}{A_{\text{cell}}} \sum_{i \in \mathcal{U}_{\text{cell}}} \sum_{j \in \mathcal{L}} \left[v_{m,n}(|\mathbf{R}_{ij} + a\mathbf{n} + s\mathbf{d}|) - v_{m,n}(|\mathbf{R}_{ij} + a\mathbf{n} + s_0\mathbf{d}|) \right]. \quad (76)$$

There is no factor one half because every cross-plane pair is counted once. In a true FCC model, two in-plane lattice indices and multiple normal layers replace the single row index. The same Mellin–Poisson strategy applies to inverse-power pair terms, but the reciprocal lattice is two-dimensional. For aluminum, a validated FCC generalized stacking-fault surface from EAM or first principles should ultimately replace the two-row energy [2].

12 Analytical Consistency Checks

The symbolic formulas admit checks that do not require inserting any parameter values:

$$\mathcal{B}_q(\delta + 1, \eta) = \mathcal{B}_q(\delta, \eta), \quad \mathcal{B}_q(-\delta, \eta) = \mathcal{B}_q(\delta, \eta), \quad (77)$$

$$\partial_\delta \mathcal{B}_q(0, \eta) = 0, \quad \Delta \mathcal{B}_q(\delta_0, \delta_0; \eta) = 0. \quad (78)$$

The direct and reciprocal forms must agree for every $q > 1$, $\eta > 0$, and real δ . At large η , the registry-dependent Fourier–Bessel terms vanish exponentially. The normal Riemann-zeta chain in Eq. (8) is a separate collinear geometry; it must not be obtained by taking $\eta \rightarrow 0$ and $\delta \rightarrow 0$ in the two-row formula, because that limit creates a coincident cross-row pair and diverges.

13 Conclusion

The exact general- m, n result is Eq. (29), and the corresponding representative energy under normal and resolved shear loading is Eq. (41). Finite-rate probability currents in the Smoluchowski equation generate dynamic loading–unloading hysteresis; interwell transitions of the unwrapped registry generate permanent reduced plastic slip. The accumulated irreversible energy is given by the nonnegative expression Eq. (63). Thus the framework can be used as a mathematically consistent ideal-slip plasticity model without fitting a sinusoidal hysteresis operator. It should not yet be presented as quantitative aluminum cyclic plasticity until the two-row LJ landscape is replaced or validated against a full FCC EAM or first-principles stacking-fault surface and the necessary dislocation hardening variables are introduced.

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